

KARTHIK LAKSHMI NARAYANA SARMA

Senior Software Engineer | AI Platform Engineering | Distributed Systems | Cloud Infrastructure

Portfolio: <https://karthiksarma.me>

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SUMMARY

Senior Software Engineer with 12+ years of experience building large-scale distributed systems, cloud infrastructure, database platforms and AI-powered developer productivity solutions. Experienced with Microsoft Azure, distributed monitoring, reliability engineering, Semantic Kernel, harness engineering, Claude Code, GitHub Copilot and AI-assisted software development.

EDUCATION

Arizona State University, Tempe, AZ, U.S.A.

Jan 2015 – Dec 2016

Master of Computer Science (MCS). Overall GPA 3.5

University Of Kerala, Trivandrum, India.

Jun 2009 – July 2013

Bachelor of Technology. Overall GPA 3.8

CORE EXPERTISE

AI Infrastructure, Harness Engineering, Agentic Systems, Distributed Computing, Platform Engineering, Cloud Architecture, Reliability Engineering, Microservices, Observability, Kubernetes, System Design.

EXPERIENCE

Senior Software Engineer – [Microsoft](#), Mountain View, California

Dec 2022 - Present

➤ OneDrive and SharePoint

- Implemented natural language to SQL query converter with Claude Sonnet LLM.
- Simplified DB Health incident troubleshooting with Claude harness.
- Reduced On-Call Engineer workload by 30% by leveraging Microsoft Semantic Kernel.
- Reduced compute utilization by introducing async invocation for managing Azure SQL pools and database operations.
- Introduced a Distributed Database Monitoring Platform for monitoring 180k databases.
- Updated Service Databases with password-less auth by leveraging Microsoft Entra-ID Auth.
- Introduced Unified Auto Heal and reduced incident mitigation time by 38%.
- Increased fault tolerance by introducing partial pool failover to geo recovery.

Software Engineer II – [Microsoft](#), Redmond, Washington

April 2020 – Dec 2022

➤ OneDrive and SharePoint

- Improved database Infrastructure management for SharePoint.
- Implemented monitoring mechanism for finding error count, reliability, latency, and API gateway errors.
- Identified/mitigated scalability issues with timer jobs which periodically monitor azure resources.
- Contributed to a domain-less reliable computational infrastructure.
- Introduced firewall handler to secure SQL servers.

Software Engineer III– [Cisco](#), San Jose, California

June 2018 – April 2020

➤ Cisco DNAC – Digital Network Architecture

- Reduced compute utilization of a real time streaming platform to analyze, manage and troubleshoot networking devices.
- Full stack development with Java 8, Jax -RS, Apache beam, gremlin tinker-pop and elastic-search.
- Developed REST API's to extract device metrics from device platform.
- identified usage statistics, enhanced streaming workflows and introduced troubleshooting mechanism with monitoring tools.
- Implemented interfaces to fetch real time issues generated by network devices.
- Proficiency with Apache Flink, Apache beam, Kubernetes, elastic-search, Java 8

Software Engineer II– [Orion Health](#), Scottsdale, Arizona

Dec 2017 – June 2018

➤ Business Intelligence & Analytics

- Enhanced medical analytics platform developed using Apache Spark Streaming, Scala, Kafka and Elastic-Search.
- Implemented ETL jobs which aggregate meaningful attributes from healthcare data.
- Built a stable platform which handles high volume of real time data and provide metrics for the end user.
- Programming in Apache Spark using Dataframes, Datasets, RDD, and DStreams.
- Well acquainted with git source control system, jira issue tracker, Jenkins pipeline and amazon web services.

Software Engineer I– [Orion Health](#), Scottsdale, Arizona

Jan 2017 – Dec 2017

➤ **Business Intelligence & Analytics**

- Contributed to a medical analytics platform developed using Apache Spark, Scala, Java 8 and Elastic-search
- Rest API development using Java 8, J2EE, JAX-RS, REST services
- Front end user interface development using React, ES6, Webpack, Redux, Javascript.
- Work closely with team on features and peer review other developer code.

Application, Demo and Development Intern – [Alcatel](#), San Diego, California

May 2016 – Aug 2016

➤ **Microservices and cloud platform**

- Developed REST'ful services in Go lang for an application in the IoT domain.
- Managed distributed cluster and helped production release via Docker image.
- Proficient in application production and deployment using Apache Mesosphere, Apache Marathon and Docker machine.
- Developed skills in Linux virtualization using QEMU, KVM, libvirt.
- Used Amazon Web Services (AWS) / Elastic Cloud Compute (EC2).

Software Engineer, [Quest Global](#), Trivandrum, Kerala

Oct 2013 – Nov 2014

➤ **Patient Information Management System, Toshiba Medical Systems Corporation**

- Contributed software engineering expertise in the development of PIMS (Patient Information Management System) through all phases of software lifecycle, from requirements definition through successful deployment.
- Provided user requirements analysis, design and programming support for enhancement of PIMS application.
- Assisted many senior developers in creating quick fixes to bugs for the software workflow process.
- Developed troubleshooting skills through log analysis and built hands-on experience in implementing design patterns.

PROGRAMMING SKILLS

- **Programming Languages** – Java, C#, Go, Python, Javascript
- **Container Orchestration Platforms:** Kubernetes, Apache Mesos, Apache Marathon, Docker
- **Big Data Frameworks:** Apache Spark, Apache Flink, Apache Beam
- **AI Frameworks:** Microsoft Semantic Kernel, Lang Chain, Pytorch